Metabolic imaging by NAD(P)H phasor FLIM to analyze impact of HER2 expression and hypoxic culture conditions across multiple cancer cell lines

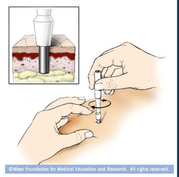
Introduction

Typically, in vitro studies of cultured cells are performed in conditions with atmospheric oxygen concentrations. However, cells in vivo experience reduced access to oxygen, especially those within a developing tumor. To mimic the hypoxic conditions of in vivo cancer cells, we used a low oxygen incubator to grow breast cancer and squamous cell carcinoma cells to image using the multiphoton laser scanning confocal LEICA microscope. To analyze the effect on NADH lifetime and NADH bound fraction in breast cancer cells, we used the human epidermal growth factor receptor 2 (HER2) inhibitor, AG825.

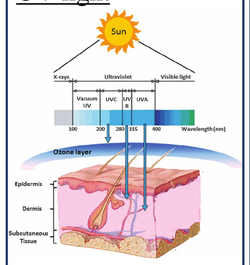
Background

1 in 5 Americans develop skin cancer
2 people die every hour in the US due to skin cancer
$8.1 billion dollars is spent every year treating skin cancer

Detection



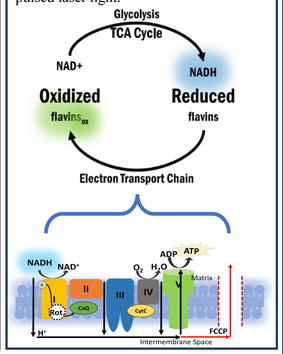
UV light



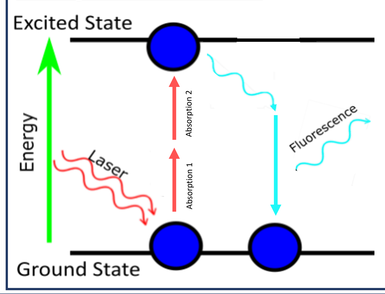
UV light can be split into wavelength ranges of UVA, UVB, and UVC. Sunlight is about half UVA, half UVB, with UVC blocked by the ozone layer. UV light causes skin cancer by both directly and indirectly damaging DNA in skin cells.

NADH

NADH is an endogenous, fluorescent indicator of metabolic activity, which can be two-photon excited using 740-nm pulsed laser light.



Multiphoton excitation



The LEICA microscope used to image MDA MB 231 and SCC 74 A/B cells uses multiphoton microscopy instead of single photon excitation to lessen the loss of light. Two photon excitation also uses longer wavelengths which creates a lower scattering coefficient.

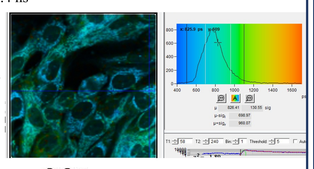
Decay Lifetimes

Free NADH: lifetime ≈ 0.4 ns

Protein bound NADH: lifetime ≈ 1-4 ns

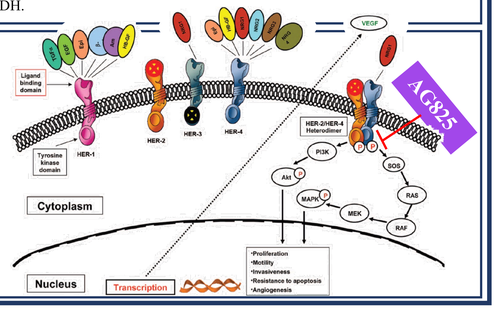


The time NADH remains in its excited state depends on whether it is free, or protein bound. A lifetime decay graph shows the ratio of bound to free NADH.



Inhibition

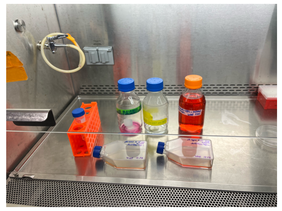
Overexpression of HER2 is associated with increased cancer aggressiveness and poor treatment prognosis. AG825 is a selective inhibitor of HER2. Mitochondrial toxins, FCCP (uncoupler) and Rotenone (inhibitor), which is often used as a rat poison, are used to assess electron transport chain activity.

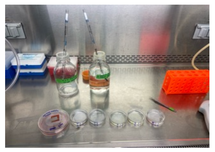


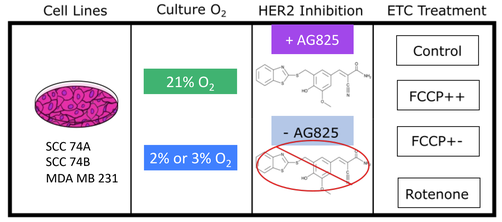
Methods and Materials

Cell culture

A typical setup for cell culture (right) included media for the cells, PBS, Trypsin, and flasks all under the fume hood. Cell culture was done twice a week for each cell line. Cells were first counted and then passed on to a new flask.







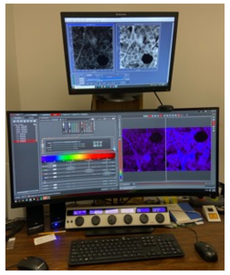
Cells were incubated (right) in either a 21% O_2 or a 2% O_2 environment. Half of the cells were treated with AG285 in culture. Prior to imaging, cells were treated with either a mitochondrial uncoupler (FCCP; with or without glucose) or a mitochondrial inhibitor (Rotenone) to assess electron transport chain activity.

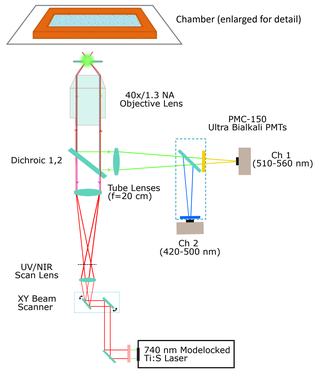


Imaging

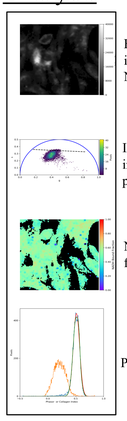


Two culture plates of cells, containing coverslips of each ETC treatment were imaged in a day using the LEICA microscope (left). Laser power was kept relatively consistent between 1.2V and 1.3V. A wavelength of 740nm was used. The SPC 64 application for imaging paired with the LAS X application for laser control and configuration were used to identify a region of cells and image a section over a 2-minute time interval.





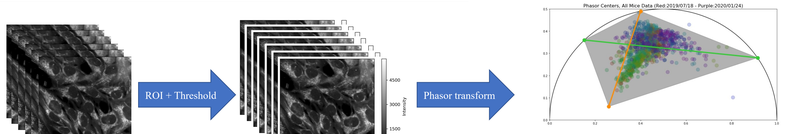
Analysis



Statistical methods:

Bound fraction/ Intensity ~
Culture Oxygenation * AG825 * Metabolic Treatments

The R programming language was used for statistical analysis. The main factors in our experimental design were oxygen culture conditions (21% and 2%), HER2 inhibition (AG825 or cell line), and metabolic treatment (4 levels). In order to establish a best fit model to determine significance of interaction terms, an AIC table was created. Using the significant interaction terms, we ran a type III factorial ANOVA.



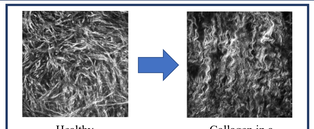
Cosine Transform $g(\omega) = \frac{\int F(x,y,t)\cos(\omega t)dt}{\int F(x,y,t)dt}$

Sine Transform $s(\omega) = \frac{\int F(x,y,t)\sin(\omega t)dt}{\int F(x,y,t)dt}$

Sine (s) and cosine (g) transforms of the fluorescence decay pixel by pixel are plotted, and the average value for each image in the study was plotted as a 2D histogram on the universal circle to form a phasor plot.

Collagen

Collagen is a structural protein found within skin that holds skin together. When cancer cells begin to grow and a tumor is formed collagen is rearranged in the skin. Using our own microscope set up to detect the forward to backward ratio of scattered light sent into a collage sample we can begin to detect these structural changes in collagen.



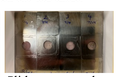
Healthy collagen

Collagen in a tumor

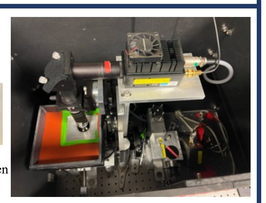
Set up and imaging:

In the machine shop, we used a milling machine to create a stand for the laser.





Slides were made from rat tail collagen and Human Type 1 collagen.



Histology



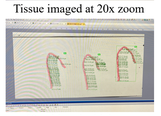


Histology was a regular task each week in which skin tissue samples from mice used previously in the lab were placed in the slide scanner in order to produce images as seen on the left to eventually analyze the epidermal length from each tissue sample. Samples were preserved in either ethanol or formalin, stained with H&E, and then scanned.

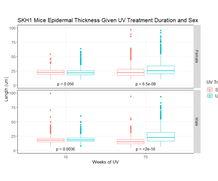
Analysis:



Tissue imaged at 20x zoom

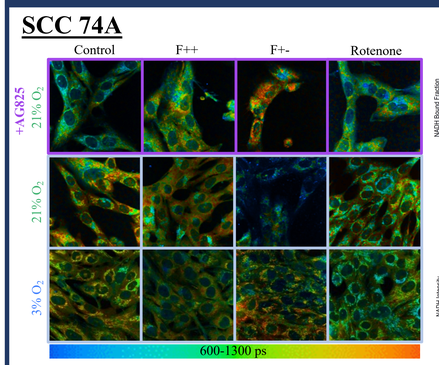


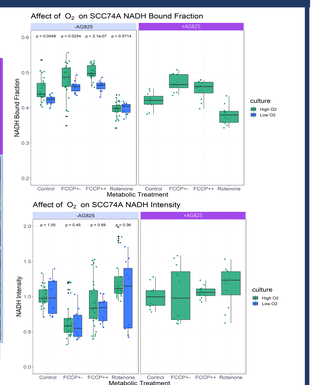
50 epidermis thickness measurements per tissue



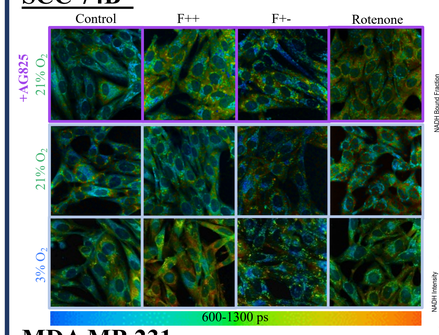
Results

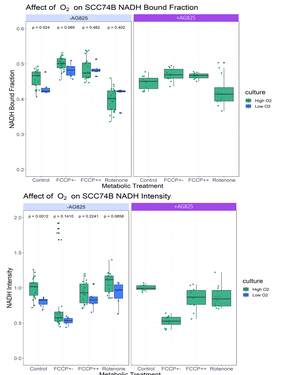
SCC 74A



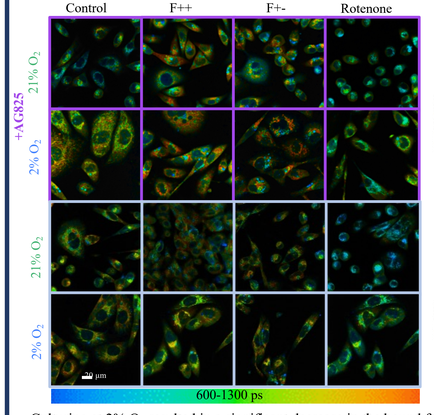


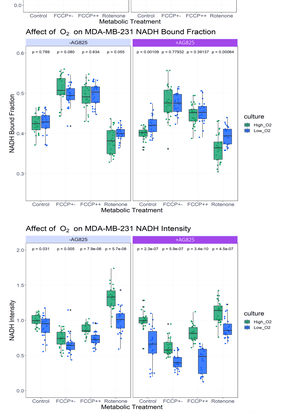
SCC 74B





MDA MB 231





Culturing at 2% O_2 resulted in a significant decrease in the bound fraction and increase in intensity of NAD(P)H. In cell-lines where HER2 was inhibited, the effect of ETC inhibition/uncoupling was decreased. The assessment of NAD(P)H successfully identified differences in metabolism linked to the environmental variables including oxygenation and HER2 expression. This implies that HER2 expression plays a significant role in the regulation of the ETC in multiple cancer cell lines in vitro.

References

Ross *et al.* (2009) The HER-2 Receptor and Breast Cancer: Ten Years of Targeted Anti–HER-2 Therapy and Personalized Medicine. Oncologist 14:320 –368

Sukstorg *et al.* (2021) Lasers and Lab Mice: Quantifying changes in cellular metabolism and structure as skin cancer develops